Explainable AI Methods in Skin Lesion Classification: A Literature Review

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Abstract

Deep learning models excel at accurately classifying melanoma but remain difficult to interpret. Saliency-based explainable AI methods such as Grad-CAM, LIME, and SHAP are widely used, but their evaluation is often qualitative or limited to simple lesion-level overlap with segmentation masks.

This literature review examines how saliency methods are assessed in dermoscopic image classification, discussing quantitative lesion-level metrics (IoU, Dice, SIR) and concept-level perspectives inspired by the ABCD dermatologists' heuristic. In particular, it considers ABCD-based pseudo-concept maps as a structured way to support explanation evaluation.

By introducing clinically structured spatial evaluation, this work aims to better align model explanations with established dermatological reasoning.

1 Introduction

Early detection of melanoma is critical, as diagnostic delay reduces significantly patient survival. Dermoscopy enables detailed visual assessment of pigmented lesions, and recent advances in deep learning have led to high-performing automated classification systems trained on large datasets such as ISIC and HAM10000. Convolutional neural networks (CNNs) now achieve strong performance in melanoma detection. Despite this progress, the decision-making process of CNNs remains opaque. In clinical settings, diagnostic decisions carry asymmetric risks: false negatives can have severe consequences and therefore predictive accuracy alone is insufficient. Explainable artificial intelligence methods (XAI) have been proposed to visualise model attention, but their evaluation is limited to qualitative inspection or coarse lesion-level overlap. This work systematically compares popular saliency-based XAI methods (Grad-CAM, LIME, SHAP) and proposes an evaluation framework extending lesion-level metrics with ABCD-inspired pseudo-concept maps derived from ground-truth segmentation masks. By integrating clinically structured regions into evaluation, this study seeks to better align model explanations with established dermatological reasoning.

2 Background and Related work

2.1 Clinical context and Diagnostic Criteria

Dermatology focuses on the diagnosis and treatment of diseases affecting the skin, including both benign conditions and malignant skin cancers. In clinical practice, the assessment of pigmented skin lesions is guided by dermoscopy, which examines lesions under magnification and controlled illumination, revealing subsurface structures invisible to the naked eye : characteristic patterns related to colour distribution, texture, and morphology. However, interpretation of dermoscopic images requires specialised training and remains subject to inter-observer variability. Public benchmarks from the International Skin Imaging Collaboration (ISIC) and the HAM10000 have played a central role in this development, providing curated dermoscopic images with diagnostic labels and, in some cases, expert-validated segmentation masks [1], [2]. Although these datasets are typically organised as multi-class classification tasks, this work adopts a melanoma versus non-melanoma formulation for consistency with common experimental protocol, while recognising that this binary mapping corresponds to the broader clinical distinction between benign and malignant lesions. Clinical assessment of pigmented skin lesions is often guided by heuristic criteria such as the ABCD rule, which evaluates lesions according to Asymmetry, Border irregularity, Colour variation, and Diameter (ABCD) and remains a widely used reference in dermoscopic practice [3]. These criteria structured early CAD systems through explicit geometric and color descriptors [4], [5]. They continue to influence contemporary AI-based dermatology, although often in an implicit manner [6], [7].

2.2 Deep Learning for Skin Lesion Classification

The emergence of convolutional neural networks (CNNs), shifted this paradigm from handcrafted feature extraction, towards end-to-end representation learning. CNNs automatically learn hierarchical visual features directly from pixel data, optimising classification performance through large scale training. Supported by benchmarks such as ISIC and HAM10000, CNNs have become the dominant paradigm for dermoscopic image classification, with common architectures such as ResNet, DenseNet, and EfficientNet [1], [2]. However, this performance-driven approach reduced the explicit modelling of clinically interpretable concepts. Evaluation in ISIC challenges is primarily reported through accuracy or AUC, which provides limited insight into the reasoning behind individual predictions. As noted in surveys of medical imaging AI, predictive performance alone is insufficient for safe clinical deployment [8]. This opacity motivates the integration of explainable AI methods to complement CNN-based classification and support clinical trust.

3 Explainable AI Methods in Medical Imaging

Deep learning models for medical imaging are often treated as black boxes, which creates concerns about transparency, safety, and clinical acceptance. Dhar et al. describe how this lack of interpretability undermines trust in automated decision systems and argue that medical AI must provide understandable, case-level explanations rather than opaque scores alone [9]. Srivastava et al. similarly emphasise that explainable artificial intelligence is essential for reliability and trustworthiness across domains such as medical imaging, financial markets, and sentiment analysis, highlighting the need for techniques that expose model reasoning while preserving performance [10].

In dermatology, recent skin lesion classification pipelines by Nigar et al. [11], Mridha et al. [12], and Metta et al. [13] integrate post-hoc explainability methods to complement convolutional neural network predictions with visual evidence that can be inspected by clinicians. A common choice in these works is Grad-CAM, introduced by Selvaraju et al. [14], which uses gradient information flowing into the final convolutional layers to generate class-specific heatmaps that highlight the regions most influential for a given prediction. When applied to dermoscopic images, Grad-CAM can reveal whether the network attends primarily to the lesion or to surrounding artefacts, providing an intuitive spatial explanation that aligns with how dermatologists visually assess lesions.

Alongside saliency-based methods, model-agnostic feature attribution techniques such as LIME and SHAP are increasingly used to probe medical classifiers. LIME, proposed by Ribeiro et al. [15], approximates the behaviour of a complex model in the neighbourhood of a single instance by fitting an interpretable surrogate, for example, a sparse linear model over super-pixels, thereby indicating which image regions contributed most to an individual prediction. SHAP builds on Shapley values from cooperative game theory to assign additive importance scores to features, and has been recognised by Srivastava et al. [10] as a key tool for quantifying feature contributions in high-stakes applications. In medical imaging, these methods are used to highlight clinically meaningful structures (such as irregular borders or atypical pigmentation patterns) and to verify whether the model's focus is consistent with established diagnostic cues, as demonstrated in the explainable skin lesion frameworks of Nigar et al. [11], Mridha et al. [12], and Metta et al. [13].

Together, LIME, SHAP, and Grad-CAM provide complementary perspectives on model behaviour: Grad-CAM focuses on where the network attends in an image, whereas LIME and SHAP offer more explicit feature-level attributions that can be related to human concepts. Building on the way these methods are deployed in existing dermatology studies, this research systematically compares them within a common experimental setting, with subsequent sections introducing quantitative metrics and ABCD-based pseudo-concept maps to assess and structure their explanations more rigorously.

4 Evaluation Framework

4.1 Ground-truth masks

Expert-validated lesion segmentation masks provided in the ISIC 2018 and HAM10000 datasets are used as spatial ground truth in this study [1], [2]. These masks delineate the lesion boundaries and define clinically relevant regions within each image. In the context of explainable AI evaluation, segmentation masks enable objective assessment of whether saliency maps concentrate within medically meaningful areas. Overlap-based metrics such as Intersection over Union (IoU), Dice coefficient, and Saliency-Inside Ratio (SIR) are computed between saliency maps and lesion masks to quantify spatial alignment. Although automated segmentation methods such as U-Net [16] and its later variants have demonstrated high agreement with expert annotations, segmentation modelling is not the focus of this work. Instead, the provided expert masks are used directly to ensure reproducible and controlled evaluation. As explored in later sections, lesion masks providing a basic definition of clinical relevance, can also serve as a spatial foundation for more fine-grained analyses of clinically inspired visual concepts within the lesion.

4.2 Lesion-Level Metrics

Evaluating explainability in medical imaging remains challenging, as explanations must be both faithful to the underlying model and meaningful for clinical decision-making. Dhar et al. highlight that many deep learning systems are still assessed through ad hoc or purely visual inspection of saliency maps, which is insufficient in safety-critical domains such as healthcare. Similarly, Srivastava et al. [10] argue that explainable AI should be supported by rigorous and measurable criteria reflecting reliability and trustworthiness, rather than relying solely on qualitative heatmap interpretation. To address this limitation, recent work in dermatology and medical imaging has adopted quantitative, mask-based evaluation strategies that compare saliency maps with expert-annotated lesion regions. In skin lesion classification, localisation-aware explainability pipelines have been explored by Nigar et al. [11] and Mridha et al. [12], where visual evidence is assessed in relation to segmented lesion areas. Building on this perspective, the present study employs three complementary spatial metrics: Intersection over Union (IoU), Dice coefficient, and Saliency-Inside Ratio (SIR). IoU and Dice measure geometric overlap between a binarised saliency map and the ground-truth lesion mask, extending classical segmentation evaluation practices to explanation assessment. SIR, in contrast, quantifies the proportion of total saliency intensity located within the lesion, providing an intensity-based concentration measure. Together, these metrics capture both spatial alignment and saliency distribution. However, spatial overlap alone does not fully capture broader aspects of trust and reliability. Dhar et al. [9] emphasise that robustness, bias, and ethical considerations also influence trust in medical AI systems. In this direction, Ieracitano et al. [17] propose TIxAI, a Trustworthiness Index for eXplainable AI in skin lesion classification, which quantifies the difference in relevance between lesion and non-lesion areas. Inspired by this multidimensional perspective, the present practicum combines overlap-based metrics with ABCD-inspired pseudo-concept maps to introduce structured clinical relevance into explanation evaluation.

4.3 Pseudo-Concepts for Explainability

Despite the strong performance of modern convolutional neural networks, their decision-making process is typically opaque. Saliency-based explainability methods provide spatial attribution maps highlighting regions that influence model predictions, but evaluation is often limited to lesion-level alignment. This implicitly assumes that all regions within the lesion are equally relevant, which does not fully reflect clinical reasoning. In practice, clinicians focus on specific visual cues, such as irregular borders or heterogeneous pigmentation, rather than treating the lesion as a homogeneous region. Concept-based explainability methods have been proposed as a way to bridge this gap by linking model behaviour to higher-level, human-interpretable concepts [19]. Importantly, such concepts need not be perfectly annotated clinical entities; approximate or heuristic definitions can still provide meaningful insight. Within this context, the ABCD criteria can be interpreted as clinically inspired pseudo-concepts that introduce structured, concept-level regions within the lesion. In this practicum, ABCD-based pseudo-concept maps are used to refine lesion-level explainability evaluation, enabling a more fine-grained and clinically grounded analysis without requiring additional expert concept annotations. Rather than modelling full clinical ABCD reasoning, this study introduces spatial proxies inspired by the ABCD criteria and integrates them into a quantitative evaluation framework for saliency methods

5 Concept-Level and Structured Explainability Methods

5.1 ABCD Framework and pseudo-concept maps

While saliency-based methods such as Grad-CAM highlight where a model attends, they provide limited insight on how or why a specific decision is made. Furthermore, their outputs, typically visual heatmaps, may be sensitive to artefacts, adversarial perturbations, and dataset biases. This raises concerns of over-confident interpretations in safety-critical domains such as medical imaging [9], [10]. Alternative explainability approaches attempt to provide richer interpretations. Example-based methods such as ABELE, explain model predictions using exemplar and counter-exemplar images rather than relying only on pixel-level saliency maps [13]. Contextual Importance and Utility (CIU) offers a different perspective on explainability by focusing on how important a feature is and how favourable its value is within a specific context[19]. Counterfactual explanations represent another promising direction, aiming to identify minimal changes to an input that would alter a model's prediction. In medical imaging, counterfactuals can help highlight decision boundaries and clarify why a particular classification was made [20]. Although conceptually powerful, these approaches often require training complex generative models, or complex adaptation to high-resolution medical images, and their evaluation is typically qualitative.

Recent work by Zheng et al. [21] proposes a cross-modal framework that aligns dermoscopic image embeddings with clinical derived ABC features. They use contrastive learning, followed by LLM-based report generation, clearly integrating clinical concepts into the model architecture. In contrast, the present work does not aim to redefine a classifier but to examine how post-hoc saliency methods in dermatological AI. It considers a structurally grounded perspective by discussing the potential use of ABCD-inspired pseudo-concept maps derived from lesion segmentation masks. It may offer a way to introduce a structured and clinically relevant evaluation without modifying the underlying model.

5.2 Limitations and Research Scope

The proposed pseudo-concept maps are heuristic spatial approximations rather than true clinical concept annotations. Their generation from segmentation masks remains technically challenging and requires clear formalisation. More advanced approaches integrate richer clinical descriptors directly into model architectures [Zheng et al., 2025; Jütte et al., 2024]. In contrast, this work emphasises the need for quantitative grounded evaluations before full clinical concept modelling.

6 Conclusion and Research Positioning

Existing work in dermatological XAI predominantly evaluates saliency maps at lesion level using overlap metrics such as IoU, Dice, and SIR. While more advanced approaches, including concept-based and counterfactual explanations, have been proposed, systematic quantitative comparisons across methods remain limited. The literature reveals a gap between lesion-level evaluation and clinically structured reasoning. Although overlap metrics provide an objective spatial baseline, they treat the lesion as a homogeneous region and may not fully reflect dermatological assessment criteria. By synthesising lesion-level metrics with concept-level perspectives inspired by the ABCD rule, this review highlights the potential role of pseudo-concept maps as a structured approach for future explainability evaluation. Introducing clinically motivated spatial structure may support a more systematic and dermatologically grounded assessment of saliency methods.

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